



Sentinel-2 L1c: Quality Processes Review

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1 Introduction

1.1 Scope

This document summarises a Data Product Quality Processes Review for the Copernicus Sentinel-2 L1b data product as part of the EODH Data Product Quality Process [AD-1]. The purpose is to demonstrate that the product has been developed and generated following processes that adhere to best practices.

To achieve this, the EODH Data Product QA Process adheres to the ESA/NASA EO Product QA Framework [RD-1]. This framework defines requirements for a variety of aspects of Data Product quality – awarding grades of *Basic*, *Good*, *Excellent*, or *Ideal*. Such assessments are reported with a maturity matrix – this is presented in Section **Error! Reference source not found.**. The rest of the document provides the underpinning evidence justifying the awarded grades. This assessment has been completed by the EODH QA Service on behalf of the Data Product provider.

The result of this evaluation, referred to as a *QA Check Result*, will be added as an annotation to the EODH Catalogue for user interaction.

1.2 Changes Log

Issue	Date	Description
0.1	25/3/2025	Initial draft for inclusion in pathfinder demonstration.

1.3 Applicable Documents

[AD-1] EODH QA Service, 'Data Product Quality Assurance Process', 2025.

1.4 Reference Documents

- [RD-1] Hunt et al., 'Earth Observation Mission Quality Assessment Framework' v2.2, EDAP Project Document, 2021. [Available Online](#).
- [RD-2] SentiWiki, see : <https://sentiwiki.copernicus.eu/web/sentinel-2>
- [RD-3] Gorroño, J. et al., *A Radiometric Uncertainty Tool for the Sentinel 2 Mission*. Remote Sens. 2017, 9, 178. <https://doi.org/10.3390/rs9020178>
- [RD-4] Copernicus Open Data Policy, [Available Online](#).
- [RD-5] CS team *Sentinel-2 Product Specification Document* v15, S2-PDGS-CS-DI-PSD, 2024. [Available Online](#).
- [RD-6] ESA CARD4L self-assessment of Sentinel-2 Surface Reflectance PFS v5, [Available Online](#)
- [RD-7] S2-MPC, *Level-1 Algorithm Theoretical Basis Document* v1.1, S2-PDGS-MPC_ATBD-L1_V1.1, 2023. [Available Online](#).
- [RD-8] E. Mazy, et al., "Sentinel-2 diffuser on-ground calibration", Proc. SPIE 8889, Sensors, Systems, and Next-Generation Satellites XVII, 88890W (24 October 2013); <https://doi.org/10.1117/12.2028854>
- [RD-9] OPT-MPC, Data Quality Report Sentinel-2 MSI L1C January 2025, OMPC.CS.DQR.002.12-2024, 2025. [Available Online](#).

- [RD-10] Bouvet, M. et al. *RadCalNet: A Radiometric Calibration Network for Earth Observing Imagers Operating in the Visible to Shortwave Infrared Spectral Range*. Remote Sens. 2019, 11, 2401. <https://doi.org/10.3390/rs11202401>
- [RD-11] Lamquin, N. et al. (2019). An inter-comparison exercise of Sentinel-2 radiometric validations assessed by independent expert groups. Remote Sens. of Env. 2019, 233, 111369. <https://doi.org/10.1016/j.rse.2019.111369>.
- [RD-12] S2-MPC, *Copernicus Sentinel-2 GRI as Database of GCPs in L1B & L1C - Product Handbook v3*, S2-MPC_PHB_GCP_L1B_L1C_GRI, 2023. [Available Online](#).
- [RD-13] S2-MPC, *Copernicus Sentinel-2 GRI as Database of GCPs in L1B & L1C - Validation Report*, S2-MPC_VAL_GCP_L1B_L1C_GRI, 2023. [Available Online](#).
- [RD-14] Gorroño, J., et al. (2018). *Providing uncertainty estimates of the Sentinel-2 top-of-atmosphere measurements for radiometric validation activities*. European Journal of Remote Sensing, 51(1), 650–666. <https://doi.org/10.1080/22797254.2018.1471739>
- [RD-15] Vera Leister-Taylor, *Copernicus Digital Elevation Model Validation Report v3*, GEO.2018-1988-2, 2020. [Available Online](#).
- [RD-16] Strobl, P. et al., *THE DIGITAL ELEVATION MODEL INTERCOMPARISON EXPERIMENT DEMIX, A COMMUNITY-BASED APPROACH AT GLOBAL DEM BENCHMARKING*, Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci., XLIII-B4-2021, 395–400, <https://doi.org/10.5194/isprs-archives-XLIII-B4-2021-395-2021>, 2021.

1.5 Acronyms & Abbreviations

EDAP	Earthnet Data Assessment Project
EO	Earth Observation
EODH	Earth Observation Data Hub
ESA	European Space Agency
NASA	National Aeronautics and Space Administration
NCEO	National Centre for Earth Observation
NPL	National Physical Laboratory
QA	Quality Assurance

2 Review Summary – Product Maturity Matrix

The *Data Product Quality Processes Review* is aimed at demonstrating that a given Data Product is developed and generated following processes that adhere to best practices.

To achieve this, the EODH *Data Product QA Process* adheres to the ESA/NASA EO Product QA Framework [RD-1]. Such assessments are reported in a maturity matrix style, where each box represents an area of assessment for which underpinning requirements are defined – the box colour represents the achieved grade, *Basic*, *Good*, *Excellent*, or *Ideal*. Additionally:

- *Not Assessed* means this aspect of data quality was not assessed at this time.
- *Not Assessable* means the assessment of this aspect of data quality was not possible due to a lack of available evidence.
- The padlock symbol means that the supporting evidence reviewed is not publicly available.

The *Data Product Quality Processes Review* for Copernicus Sentinel-2 L1c is presented in Figure 1. The rest of the document provides the underpinning evidence justifying the awarded grades, as follows:

- Section 3 provides the evidence for the Product Information column.
- Section 4 provides the evidence for the Metrology column.
- Section 5 provides the evidence for the Product Generation column.

Data Product Quality Processes Review		
Product Information	Metrology	Product Generation
Product Details	Sensor Calibration & Characterisation	Calibration Algorithm
Availability & Accessibility	Geometric Calibration & Characterisation	Geometric Processing
Product Format, Flags & Metadata	Metrological Traceability Documentation	Mission-Specific Processing
User Documentation	Uncertainty Characterisation	
	Ancillary Data	

Key

- Not Assessed
- Not Assessable
- Basic
- Good
- Excellent
- Ideal
-  Not Public

Figure 1 - Data Product Quality Processes Review Maturity Matrix for Copernicus Sentinel-2 L1c

3 Product Information

Product Details	
Grade: Ideal	
Justification	Comprehensive product details provided on the Copernicus <i>SentiWiki</i> and <i>Sentinel-2 Documentation Library</i> [RD-2].
Product Name	Sentinel-2A/B MSI Level 1c
Sensor Name	MSI (Multi-spectral imager)
Sensor Type	Optical – Multichannel spectrometer (13 bands)
Mission Type	Constellation – 2 satellites
Mission Orbit	Sun Synchronous – 10:30 local time at descending node
Product Version Number	Identified by processing baseline version (currently >03), included in product name
Product ID	Products are named with following ID: MMM_MSIL1C_YYYYMMDDHHMMSS_Nxxyy_ROOO_Txxxxx_<Product Discriminator>.SAFE • first date (YYYYMMDDHHMMSS) is the datatake start sensing time. • <Product Discriminator> field is used to distinguish between different end user products from the same datatake (15 characters). • MMM: is the mission ID (S2A/S2B) • MSIL1C denotes the Level-1C product level/ MSIL2A denotes the Level-2A product level • Nxxyy: the Processing Baseline number (e.g. N0204) • ROOO: Relative Orbit number (R001 - R143) • Txxxxx: Tile Number field • SAFE: Product Format (Standard Archive Format for Europe)
Processing level of product	Level 1c – orthorectified top-of-atmosphere reflectance
Measured Quantity Name	TOA Reflectance
Measured Quantity Units	[-]
Stated Measurement Quality	Per pixel radiometric uncertainty evaluated by user software tool [RD-3]
Spatial Resolution	GSD 10 m, 20 m, and 60 m (band dependant) over 290 km swath
Spatial Coverage	The Sentinel-2 mission provides systematic coverage over the following: • continental land surfaces (including inland waters) between latitudes 56° S and 82.8° N • all coastal waters up to 20 km from the shore • all islands greater than 100 km² • all EU islands • the Mediterranean Sea • all closed seas (e.g. Caspian Sea).

Temporal Resolution	5 days revisit time at equator (combined S2A/S2B)
Temporal Coverage	• Sentinel-2A launched 2015 to present • Sentinel-2B launched 2017 to present
Point of Contact	Contact via ESA
Product locator (DOI/URL)	Available via the Copernicus Data Space Ecosystem (amongst others)
Conditions for access and use	Free, full, and open
Limitations on public access	None
Product Abstract	Extract from Copernicus <i>SentiWiki</i>: <i>Sentinel-2 is a European wide-swath, high-resolution, multi-spectral imaging mission. The full mission specification of the twin satellites flying in the same orbit but phased at 180°, is designed to give a high revisit frequency of 5 days at the Equator.</i> <i>Each of the satellites in the Sentinel-2 mission carries a single payload: the optical Multi-Spectral Instrument (MSI) that samples 13 spectral bands: four bands at 10 m, six bands at 20 m and three bands at 60 m spatial resolution. The orbital swath width is 290 km.</i> <i>The Level-1C product provides Top Of Atmosphere (TOA) reflectance images, derived from associated Level-1B products.</i> <i>The Level-1C product is composed of 110 km x 110 km tiles (ortho-images in UTM/WGS84 projection). Earth is subdivided on a predefined set of tiles, defined in UTM/WGS84 projection and using a 100 km step. However, each tile has a surface of 110 km x 110 km in order to provide large overlap with the neighbouring.</i>

Availability & Accessibility	
Grade: Ideal	
Justification	• Findable – data systematically made available through the Copernicus Data Space Ecosystem to all users. Accessible through the openEO API. • Accessibility – distributed freely under open data policy. • Interoperability – provided in standard SAFE format. • Reusability – rich product metadata enables application in a wide variety of different domains.
Compliant with FAIR principles	Yes
Data Management Plan	Copernicus Open Data Policy [RD-4]
Availability Status	Free, full, and open

Product Format	
Grade: Ideal	
Justification	Product provided in standard SAFE format [RD-5]. CEOS-ARD approved at L2 (threshold).
Product File Format	SAFE
Metadata Conventions	Includes: • SAFE format • INSPIRE • ISO 19115
Analysis Ready Data?	Yes (at L2a) [RD-6]

User Documentation		
Grade: Ideal		
Justification	Comprehensive user documentation provided on the Copernicus <i>SentiWiki</i> and <i>Sentinel-2 Documentation Library</i> [RD-2].	
<i>Document</i>	<i>Reference</i>	<i>QA4ECV Compliant</i>
Product User Guide	<i>SentiWiki</i> [RD-2]	Yes
ATBD	Level-1 Algorithm Theoretical Basis Document [RD-7]	Yes
Product Specifications	Product Specification Document [RD-5]	Yes

4 Metrology

Sensor Calibration & Characterisation	
Grade: Excellent	
Justification	Some literature exists describing the pre-flight calibration activity for Sentinel-2 (e.g., the diffuser characterisation [RD-8]), although it is not summarised into a single accessible reference. Detailed on-orbit data quality analysis is reported quarterly (and summarised annually) by the Copernicus Optical Mission Performance Centre (OPT-MPC). Analysis combines multiple methodologies and references data sources. This includes comparison to other satellites over PICS (Landsat) and uncertainty-quantified comparison to SI-traceable RadCalNet. Although RadCalNet uncertainties are comparable to the mission's requirements (5%), they are higher than the uncertainty value typically quoted by Sentinel-2 radiometric uncertainty tool (S2-RUT), so validation is not possible at this level (~2%). Comparisons primarily limited to desert sites (apart from Rayleigh scattering modelling), so doesn't cover the full range of sensor behaviour across the dynamic range. Data quality is independently monitored by the Sentinel-2 Quality Working Group (S2-QWG). There is a rich literature of calibration validation analysis from this team and more broadly (e.g., [RD-11]).
References	• Pre-flight Characterisation of the Diffuser [RD-8] • Quarterly Data Quality Report, e.g., [RD-9] • RadCalNet [RD-10] • Validation Method Intercomparison [RD-11]

Geometric Calibration & Characterisation	
Grade: Excellent	
Justification	Pre-flight and on-orbit calibration activities are organised as above for radiometric performance monitoring – including quarterly reporting. Validation is performed globally vs. the Sentinel-2 GRI (global reference image) [RD-12]. This is not analysed in terms of uncertainty, but performance metrics like CE90 [RD-13].
References	• Quarterly Data Quality Report, e.g., [RD-9] • Sentinel-2 GRI Product Handbook [RD-12] • Sentinel-2 GRI Product Validation Report [RD-13]

Metrological Traceability Documentation	
Grade: Excellent	
Justification	Sentinel-2A's TOA reflectance calibration is traceable to its on-board calibration device (solar diffuser). The calibration of this solar diffuser is described in [RD-8]. Sentinel-2B has been cross-calibrated to Sentinel-2A. This is described in the quarterly data quality reports.
References	• Pre-flight Characterisation of the Diffuser [RD-8] • Quarterly Data Quality Report, e.g., [RD-9]

Uncertainty Characterisation	
Grade: Excellent	
Justification	Sentinel-2 is a leading mission in terms of providing the users to means to access per pixel uncertainty estimates via the Sentinel-2 Radiometric Uncertainty Tool (S2-RUT) [RD-3]. The uncertainty contributors considered are quite comprehensive, although handling of some terms such as orthorectification requires further work. Some advice has been provided on how to interpret the different uncertainty contributors to the L1c uncertainty in terms of their spatial/spectral error-correlation for AOI averaging [RD-14], however this is not generalising into the tool.
References	• Sentinel-2 Radiometric Uncertainty Tool [RD-3] • Sentinel-2 Uncertainty Estimates for AOI [RD-14]

Ancillary Data	
Grade: Excellent	
Justification	The Sentinel-2 GRI and the Copernicus DEM are used for refinement of product geolocation. These are high-quality datasets. Their performance is not analysed in terms of a comprehensive uncertainty analysis, but with performance metrics like CE90 vs. references [RD-13][RD-15]. Copernicus DEM performance has been comprehensively benchmarked in the CEOS DEMIX (DEM Intercomparison Exercise), and found to be internationally leading [RD-16]. The Copernicus DEM is also used as an ancillary data source to perform orthorectification.
References	• Sentinel-2 GRI Product Validation Report [RD-13] • Copernicus DEM Validation Report [RD-15] • DEMIX [RD-16]

5 Product Generation

Calibration Algorithm	
Grade: Ideal	
Justification	Comprehensive ATBD documentation available, describing radiometric calibration algorithm. Field leading implementation.
References	Level-1 Algorithm Theoretical Basis Document [RD-7]

Geometric Processing	
Grade: Ideal	
Justification	Comprehensive ATBD documentation available, describing geometric processing algorithm. Field leading implementation.
References	Level-1 Algorithm Theoretical Basis Document [RD-7]

Mission-Specific Processing	
Grade: Ideal	
<i>Mask Generation</i>	
Justification	Comprehensive ATBD documentation available, describing generation of cloud masking algorithm. Rigorous approach taken.
Reference	Level-1 Algorithm Theoretical Basis Document [RD-7]